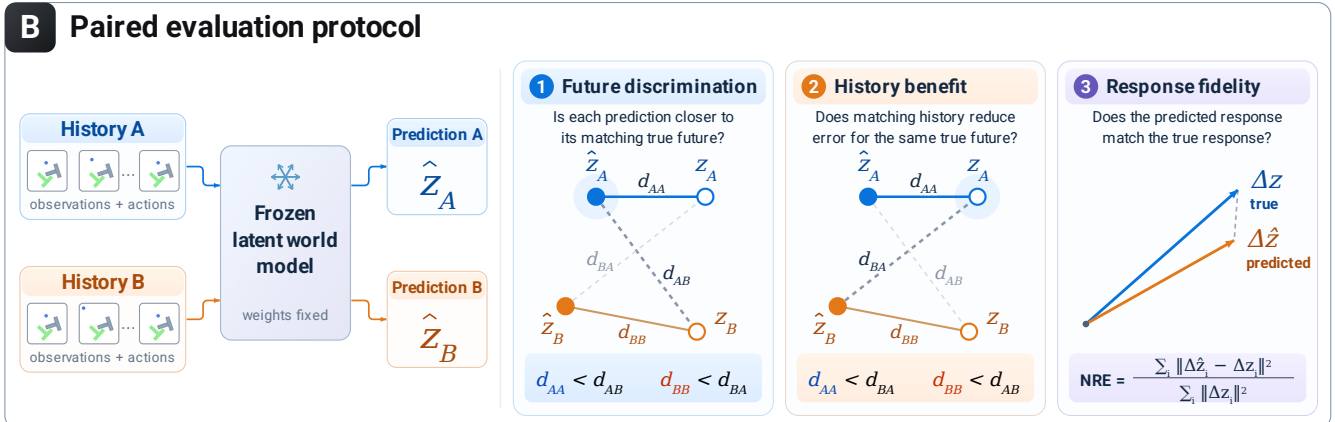

B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen latent world model
weights fixed
Prediction A
$\hat{z}_A$
Prediction B
$\hat{z}_B$
1 Future discrimination
Is each prediction closer to its matching true future?
$\hat{z}_A$
z_A
d_{AA}
d_{BA}
d_{AB}
$\hat{z}_B$
z_B
d_{BB}
$d_{AA} < d_{AB}$
$d_{BB} < d_{BA}$
2 History benefit
Does matching history reduce error for the same true future?
$\hat{z}_A$
z_A
d_{AA}
d_{BA}
d_{AB}
$\hat{z}_B$
z_B
d_{BB}
$d_{AA} < d_{BA}$
$d_{BB} < d_{AB}$
3 Response fidelity
Does the predicted response match the true response?
Δz
true
$\Delta \hat{z}$
predicted
NRE = $\frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}$